

Foundations of Artificial Intelligence & Machine Learning

Author: Andrew S. Tanenbaum | Category: Technology

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Chapter 1: Intelligence & Search

State space search: A* search algorithm and heuristic design.

Game playing: Minimax algorithm with alpha-beta pruning.

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Chapter 2: Supervised Learning

Linear regression, logistic regression, and cost minimization.

Decision trees, random forests, and gradient boosting mechanisms.

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Chapter 3: Neural Networks

Perceptrons, backpropagation, and activation functions.

Convolutional networks for computer vision and image classification.

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Chapter 4: Transformer Architecture

Self-attention mechanism and encoder-decoder foundations.

Pre-training, fine-tuning, and modern Large Language Models.

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Chapter 5: AI Ethics & Alignment

Fairness, bias mitigation, and safety in autonomous agents.

Responsible AI engineering in healthcare and education systems.